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CHAPTER 1

INTRODUCTION

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WHY DO WE NEED ARTIFICIAL INTELLIGENCE?

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- Humans are called **Homo sapiens (Wise Humans)** because intelligence is our most distinctive capability.
- For thousands of years, humans have tried to understand:
  - How we think
  - How we learn
  - How we make decisions
  - How we interact with the world
- The human brain can:
  - Perceive
  - Understand
  - Predict
  - Manipulate complex environments

Human Brain → Thinking → Learning → Decision Making → Action

Can a machine think and make decisions like humans?

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What is Human intelligence?

- It's a composition of abilities like



Learning



Reasoning



Perceiving



Understanding of Language



Feeling

Ravi Kumar B N, Asst.Prof,CSE,BMSIT

Intelligent behaviors

- **Everyday tasks**: recognize a friend, recognize who is calling, translate from one language to another, interpret a photograph, talk, cook a dinner
- **Formal tasks**: prove a logic theorem, geometry, calculus, play chess, checkers, or Go
- **Expert tasks**: engineering design, medical designers, financial analysis

Intelligence is:

- The ability to reason
- The ability to understand
- The ability to create. . .

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WHAT IS ARTIFICIAL INTELLIGENCE?

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Artificial Intelligence (AI) is the science and engineering of building intelligent systems that can perceive, reason, learn, and act effectively in different situations.

The theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

Objectives of AI

- ✅ Understand human intelligence
- ✅ Build intelligent machines
- ✅ Enable autonomous decision-making
- ✅ Solve complex real-world problems
- ✅ Act safely and efficiently in novel situations

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What is *artificial* intelligence?

human-like *vs.* rational

thought *vs.* behavior		
	Systems that think like humans	Systems that think rationally
	Systems that act like humans	Systems that act rationally

WHAT IS AI?

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

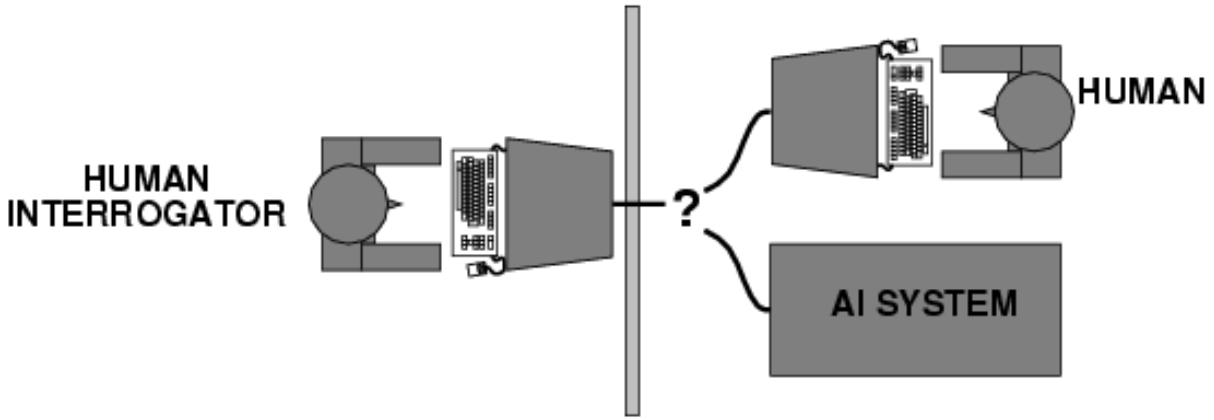
CONTD..

- Eight definitions of AI, laid out along two dimensions.
- Definitions on:
  - top are concerned with thought processes.
  - bottom definitions addresses behavior.
  - on the left measure success in terms of fidelity to human performance.
  - on the right measure against an ideal performance measure, called rationality

- A system is rational if it does the “right thing,” given what it knows.
- All four approaches to AI have been followed, each by different people with different methods.

1. ACTING HUMANLY: THE TURING TEST APPROACH

- Acting humanly means acting like a person.
- The Turing Test, proposed by Alan Turing (1950)
- A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.
- Turing deserves credit for designing a test that remains relevant 60 years later.



CONTD..

- Programming a computer to pass a rigorously applied test provides plenty to work on. The computer would need to possess the following capabilities:

NATURAL LANGUAGE PROCESSING
- **natural language processing** to enable it to communicate successfully in English;

KNOWLEDGE REPRESENTATION
- **knowledge representation** to store what it knows or hears;

AUTOMATED REASONING
- **automated reasoning** to use the stored information to answer questions and to draw new conclusions;

MACHINE LEARNING
- **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

EXAMPLE

- You are at a coffee shop, and you are waiting in line to order. The person in front of you spills their coffee all over themselves. You could simply ignore them and continue waiting in line. However, this would be insensitive. Instead, you offer to help them clean up.
- This is an example of acting humanly because it shows that you are concerned about the well-being of others. You are willing to go out of your way to help someone in need, even if you don't know them.

TOTAL TURING TEST

- Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because physical simulation of a person is unnecessary for intelligence.
- Total Turing Test includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects "through the hatch."
- To pass the total Turing Test, the computer will need:
  - computer vision to perceive objects, and
  - robotics to manipulate objects and move about

2. THINKING HUMANLY: THE COGNITIVE MODELING APPROACH

- To say program thinks like a human, we must have some way of determining how humans think.
- Need to get inside the actual workings of human minds.
- Three ways to do this:

i. **through introspection—trying to catch our own thoughts as they go by;**

ii. through psychological experiments—observing a person in action;

iii. through brain imaging—observing the brain in action.

EXAMPLE-1

- You are at a park with your child, and they see a dog running towards them. You immediately feel a sense of panic and start to run towards your child. However, you then realize that the dog is wagging its tail and seems friendly. You stop running and let the dog sniff your child.
- This is an example of thinking humanly because it involves both emotion and reason. You initially feel a sense of panic because you are worried about your child's safety. However, you are able to quickly assess the situation and realize that the dog is not a threat. You then make the rational decision to let the dog sniff your child.

EXAMPLE-2

- Another example of thinking humanly is when you are having a conversation with someone.
- You are able to listen to what the other person is saying and understand their perspective.
- You are also able to think about your own response and how it will affect the other person.
- This ability to think about both your own and the other person's thoughts and feelings is a key part of human thinking.

OTHER EXAMPLES

- Deciding which route to take to work based on traffic conditions and weather.
- Choosing what to wear for the day based on your schedule and activities.
- Deciding how to respond to a difficult email from a coworker.
- Helping your child with their homework.
- Planning a vacation with your family.

Thinking humanly is a complex process that involves both emotion and reason. It is also a process that is constantly evolving as we learn and grow.

CONTD..

- Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program.
- If the program's input–output behavior matches corresponding human behavior, that is evidence that some of the program's mechanisms could also be operating in humans.

For example, Allen Newell and Herbert Simon, who developed GPS, the "General Problem Solver" (1961), were not content merely to have their program solve problems correctly. They were more concerned with comparing the trace of its reasoning steps to traces of human subjects solving the same problems.

3. THINKING RATIONALLY: THE “LAWS OF THOUGHT” APPROACH

- Use of logic.
- The Greek philosopher Aristotle was one of the first to attempt to codify “right thinking,” that is, irrefutable(impossible to deny) reasoning processes.
- His syllogisms provided patterns for argument structures that always yielded correct conclusions.

CONTD..

- **Syllogisms :**

An instance of a form of reasoning in which a conclusion is drawn from two given or assumed propositions (premises); a common or middle term is present in the two premises but not in the conclusion.

Ex: All dogs are animals;

All animals have four legs;

Therefore all dogs have four legs.

- These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

EXAMPLE-1

- You are walking down the street, and you see a group of people gathered around a burning car.
- Your first instinct might be to run over and help, but it is important to think rationally in this situation. The fire could be dangerous, and there could be other hazards involved.
- Instead of rushing over to the scene, you should first assess the situation from a safe distance. Is the fire under control?
- Are there any people or animals in immediate danger? Once you have a better understanding of the situation, you can decide how to proceed safely.

- If the fire is small and contained, you might be able to put it out yourself using a fire extinguisher. However, if the fire is large or spreading, you should call the fire department immediately.
- It is also important to be aware of your own safety. If you are not sure whether it is safe to approach the burning car, it is best to err on the side of caution and stay back.
- This is an example of thinking rationally in real time. You are faced with a situation that is potentially dangerous, and you need to make a decision quickly. By taking the time to assess the situation and consider the potential risks and benefits of your options, you can make the most rational decision possible.

EXAMPLE-2

- You are at a job interview, and the interviewer asks you a question that you don't know the answer to. It is tempting to try to bluff your way through the answer, but this is not the rational approach. If the interviewer catches you bluffing, it will reflect poorly on you.
- Instead, the rational approach is to be honest and tell the interviewer that you don't know the answer. You can then offer to research the question and get back to the interviewer. This shows that you are honest and that you are willing to learn.
- Thinking rationally in real time can be challenging, but it is an important skill to develop. By taking the time to assess the situation and consider your options, you can make the best decisions possible, even in stressful situations.

- Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them.
- By 1965, programs existed that could, in principle, solve any solvable problem described in logical notation. (Although if no solution exists, the program might loop forever.)
- The so-called logicist tradition within artificial intelligence hopes to build on such programs to create intelligent systems

CONTD..

There are two main obstacles to this approach:

- It is not easy to take informal knowledge and state it in the formal terms required by logical notation.
- There is a big difference between solving a problem “in principle” and solving it in practice. Even problems with just a few hundred facts can exhaust the computational resources of any computer unless it has some guidance as to which reasoning steps to try first.

4. ACTING RATIONALLY: THE RATIONAL AGENT APPROACH

- An agent is just something that acts (agent comes from the Latin agere, to do).
- An intelligent agent is one that acts rationally with respect to its goals.

Ex: **An agent that is designed to play a game should make moves that increase its chances of winning the game.**

- All computer programs do something, but computer agents are expected to do more:
  - operate autonomously
  - perceive their environment
  - persist over a prolonged time period
  - adapt to change
  - create and pursue goals.

A rational agent is one that acts so as to achieve the best outcome or, when here is uncertainty, the best expected outcome.

EXAMPLE-1

- **Example**. Assume I don't like to get wet (my goal), so I bring an umbrella (my action). Do I behave rationally?
  - The answer is dependent on my knowledge and belief
  - If I've heard the forecast for rain and I believe it, then bringing the umbrella is rational.
  - If I've not heard the forecast for rain and I do not believe that it is going to rain, then bringing the umbrella is not rational.

EXAMPLE-2

You are driving to work and you come to a red light. There is no cross traffic, and you can see that the light is about to turn green.

- You could try to run the red light, but that would be irrational. There is a high risk of getting a ticket or causing an accident.
- Instead, you wait for the light to turn green. This is the rational action because it is the most likely to lead to your desired outcome (arriving at work safely and on time).

EXAMPLE-3

You are at a grocery store, trying to decide which brand of cereal to buy-

- Have two choices:

  -a generic brand that is much cheaper,

  -a brand name cereal that is more expensive.

- You could buy the brand name cereal because you are familiar with it and you like the taste. However, this would not be the rational action because it is not the most cost-effective option.
- Instead, you should buy the generic brand cereal, because it is just as good as the brand name cereal, but it is much cheaper.

The rational action is the action that is most likely to lead to the desired outcome.

- In the first example, the desired outcome is arriving at work safely and on time.
- In the second example, the desired outcome is buying the best possible value for your money.

- It is important to note that acting rationally does not always mean making the "best" decision in the abstract sense.
- Sometimes, the rational action is to make a sacrifice or compromise in order to achieve a more important goal.

For example-

you might choose to work long hours at a job that you don't love in order to save up money to buy a house. This would be a rational decision, even though it is n

- **Ultimately, acting rationally means making decisions that are based on logic and evidence, rather than on emotions or biases. It is about choosing the actions that are most likely to lead to your desired outcomes to the most enjoyable option.**

CONTD..

- In the “laws of thought” approach to AI, the emphasis was on correct inferences.
- Making correct inferences is sometimes part of being a rational agent, because one way to act rationally is to reason logically to the conclusion that a given action will achieve one’s goals and then to act on that conclusion.
- On the other hand, correct inference is not all of rationality; in some situations, there is no provably correct thing to do, but something must still be done.

- By distinguishing between *human* and *rational* behavior, we are not suggesting that humans are necessarily "irrational" in the sense of "emotionally unstable" or "insane."

One merely need note that we are not perfect:

- Not all chess players are grandmasters;
- unfortunately, not everyone gets an A on the exam.

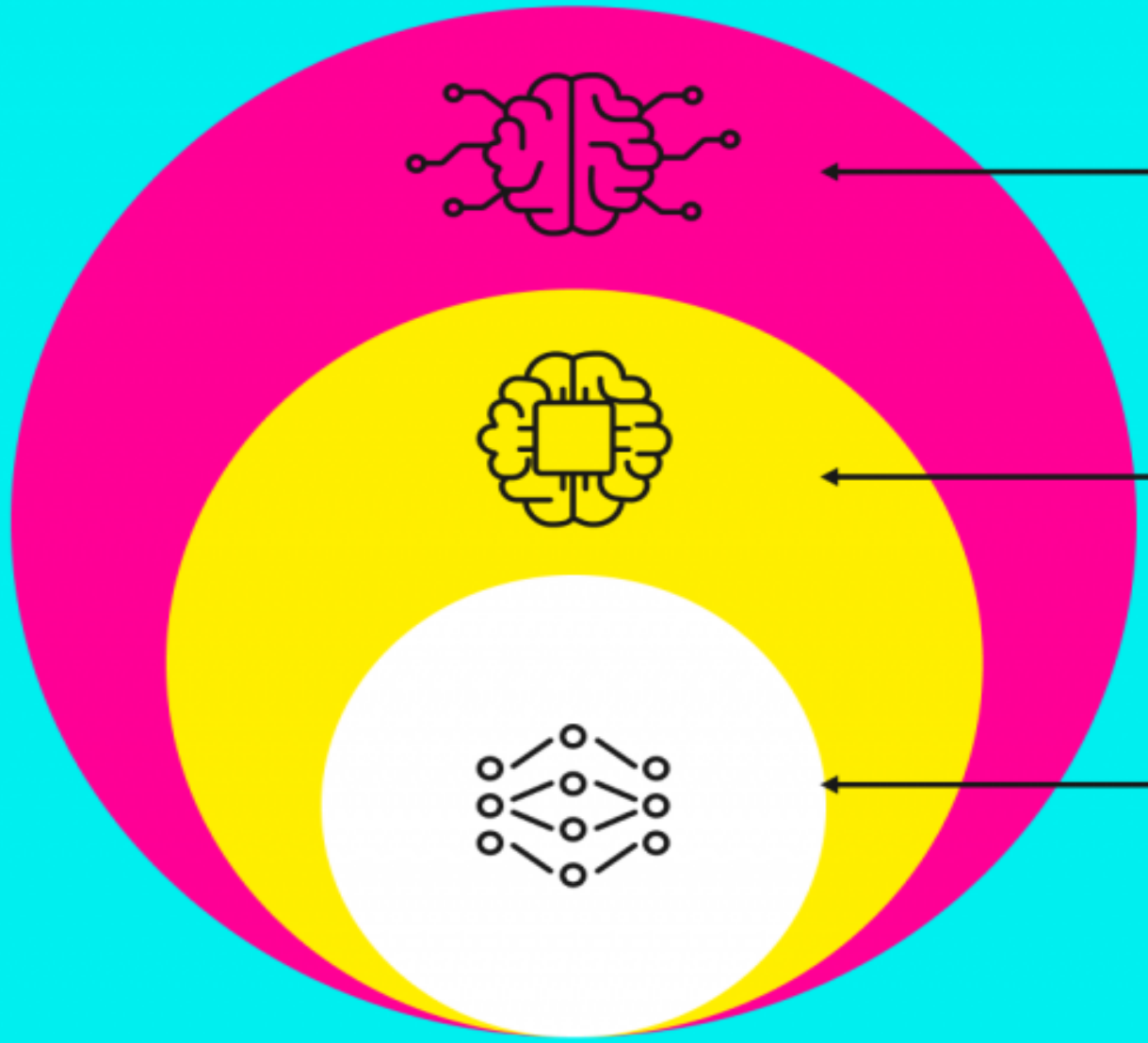
Some systematic errors in human reasoning.

A rationalist approach involves a combination of mathematics and engineering.

SOME OF AI APPLICATIONS -

- Maps and Navigation
- Facial Detection and Recognition
- Text Editors or Autocorrect
- Search and Recommendation Algorithms
- Chatbots
- Self-driving vehicles

Know the difference: AI, ML & DL



Artificial Intelligence (AI)

Any process where we use computers to mimic the cognitive abilities and functions of humans.

Machine Learning (ML)

A subset of AI. It focuses on the ability of machines to receive data and improve at a task with experience.

Deep Learning (DL)

A subset of ML. It works with multi-layered neural networks that can learn by themselves from vast amounts of data.

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- **Artificial Intelligence (AI)**: Developing machines to mimic human intelligence and behavior.
- **Machine Learning (ML)**: Algorithms that learn from structured data to predict outputs and discover patterns in that data.
- **Deep Learning (DL)**: Algorithms based on highly complex neural networks that mimic the way a human brain works to detect patterns in large unstructured data sets.

AI VERSUS TRADITIONAL PROGRAMMING

Traditional programming uses human-written rules and data to produce specific answers, while artificial intelligence (AI) uses data and known answers to figure out the rules.

Core Differences

- **Approach:** Traditional programming is rule-based, requiring developers to write explicit, step-by-step logic. AI is data-driven, using machine learning to learn patterns automatically

- **Inputs and Outputs:** Traditional software takes data and rules as input to output an answer. AI takes data and expected answers (labels) to build the rules.

- **Adaptability:** Traditional programs are static and require manual code updates to handle new scenarios. AI systems adapt dynamically and improve continuously as they process new data.

- **Predictability:** Traditional programming delivers deterministic, consistent results. AI provides probabilistic predictions that can vary or evolve based on learned patterns.

- **Transparency:** Traditional code is fully transparent and easy to debug. Complex AI models often act as "black boxes," making it harder to trace exact decision paths

NATURAL AND ARTIFICIAL INTELLIGENCE

Attributes	Natural Intelligence (Human)	Artificial Intelligence (Machine)
The ability to use sensors (eyes, ears, touch, smell)	High	Low
The ability to be creative and imaginative	High	Low
The ability to learn from experience	High	Low
The ability to be adaptive	High	Low
The ability to afford the cost of acquiring intelligence	High	Low
The ability to use a variety of information sources	High	High
The ability to acquire a large amount of external information	High	High
The ability to make complex calculations	Low	High
The ability to transfer information	Low	High
The ability to make a series of calculations rapidly and accurately	Low	High

EVOLUTION AND HISTORY OF AI

AI has evolved from early 1950s theoretical concepts into a transformative global technology through distinct eras of breakthroughs and setbacks.

The Dawn of AI (1950s–1960s)

- **1950:** Alan Turing published his landmark paper introducing the Turing Test to evaluate machine intelligence.
- **1956:** John McCarthy coined the term "Artificial Intelligence" at the Dartmouth Conference, founding AI as a formal academic discipline.
- **Early Focus:** Researchers built symbolic, rule-based systems designed to mimic human logic and solve math proofs.

Setbacks and the "AI Winters" (1970s–1980s)

- **Expert Systems:** Programs like MYCIN tried to replicate human decisions using strict rule bases in specific fields.
- **Stagnation:** Low computing power and unfulfilled promises caused funding cuts and major drops in research interest, known as AI winters

Machine Learning and Data Shift (1990s)

- **Data-Driven Learning:** Algorithms began learning patterns from data instead of relying entirely on fixed human-coded rules.
- **1997:** IBM’s Deep Blue defeated world chess champion Garry Kasparov, proving computers could master complex intellectual games.

The Deep Learning Revolution (2000s–2010s)

- **Neural Networks:** Multi-layer neural networks processed massive datasets to achieve breakthroughs in image and speech recognition.
- **2011:** IBM's Watson won *Jeopardy!*, showcasing advanced natural language processing.
- **2017:** Google researchers introduced the Transformer architecture, forming the foundation for modern language models.

Generative AI and Modern Era (2020s)

- **Foundation Models:** Large-scale transformer models like the GPT Series emerged, capable of generating human-like text, art, and code.
- **Current Focus:** The industry continues shifting toward multimodal capabilities and autonomous agentic systems.

FOUNDATIONS OF AI

The foundations of artificial intelligence (AI) rest on core academic disciplines, mathematical principles, and the concept of rational agents that perceive and act in an environment.

Core Disciplines

- **Philosophy:** Explores formal rules for the mind, how knowledge originates, and how reasoning leads to action.
- **Mathematics:** Provides logic, probability, calculus, and algorithms to handle uncertainty and make predictions.
- **Neuroscience:** Inspires artificial neural networks by studying how biological brains process information.
- **Computer Science:** Supplies the hardware infrastructure, data structures, and programming languages required to run AI models.
- **Psychology & Cognitive Science:** Investigates how humans think, act, and solve problems to model intelligent behavior.
- **Linguistics:** Powers natural language processing (NLP) for human-computer communication.
- **Economics:** Defines decision-making strategies to maximize utility and payoffs.

Key Technical Pillars

- **Machine Learning (ML):** Systems learn patterns from large datasets rather than relying on explicit programming.
- **Intelligent Agents:** Entities that use sensors to perceive surroundings and actuators to take successful actions.
- **Knowledge Representation:** Storing information via logic and semantic networks so computers can reason and infer solutions.

TYPES OF AI

Artificial intelligence is grouped into categories based on its capabilities and how it functions.

Types Based on Capabilities (Levels of Intelligence)

- ✓ **Narrow AI (Weak AI):** This AI does one specific job. It powers tools we use today, like voice assistants and facial recognition. It cannot work outside its programmed tasks.
- ✓ **General AI (Strong AI):** This theoretical AI would match human intelligence across any task. It could learn and adapt like a person. It does not exist yet.
- ✓ **Super AI (Superintelligence):** This is a future concept where machines surpass human intelligence in every way. It remains purely hypothetical.

Types Based on Functionality (How They Work)

- ✓ **Reactive Machines:** These basic systems have no memory. They react to current inputs using pre-set rules. An example is a chess computer like Deep Blue.
- ✓ **Limited Memory AI:** These systems use past data to learn and make decisions. Most modern AI, like self-driving cars, belongs here.
- ✓ **Theory of Mind:** This future AI would understand human emotions, beliefs, and needs. It is still in the research phase.
- ✓ **Self-Aware AI:** This ultimate stage describes an AI with its own consciousness and self-awareness. It exists only in theory.

APPLICATIONS OF AI IN ENGINEERING AND SOCIETY

Artificial intelligence transforms engineering and society by automating complex tasks, improving efficiency, and driving major innovations

Applications in Engineering

- **Generative Design:** Algorithms create thousands of optimal product options based on set rules, cutting down material waste and weight. Platforms like Neural Concept help engineers test designs faster.
- **Predictive Maintenance:** Machine learning models track equipment temperature and vibration to spot failures before they happen, which saves money and avoids downtime.
- **Smart Manufacturing:** Computer vision cameras inspect factory parts for tiny defects to improve quality control.
- **Fast Simulations:** Physics-aware AI predicts stress points and fluid dynamics in seconds instead of taking days with standard software

Applications in Society

- **Healthcare:** AI tools analyze patient records, assist with medical imaging diagnostics, and support early disease detection.
- **Autonomous Vehicles:** Self-driving cars and delivery drones use sensors and computer vision to navigate roads and avoid hazards safely.
- **Daily Automation:** Virtual assistants and smart home systems manage energy use, translate languages, and handle daily scheduling.
- **Sustainability:** Smart grids and algorithms optimize urban power consumption and reduce industrial resource waste.

CHAPTER 2

INTELLIGENT AGENTS

AGENTS AND ENVIRONMENTS

An agent is anything that perceives its environment through **sensors** and acts upon that environment through **actuators**

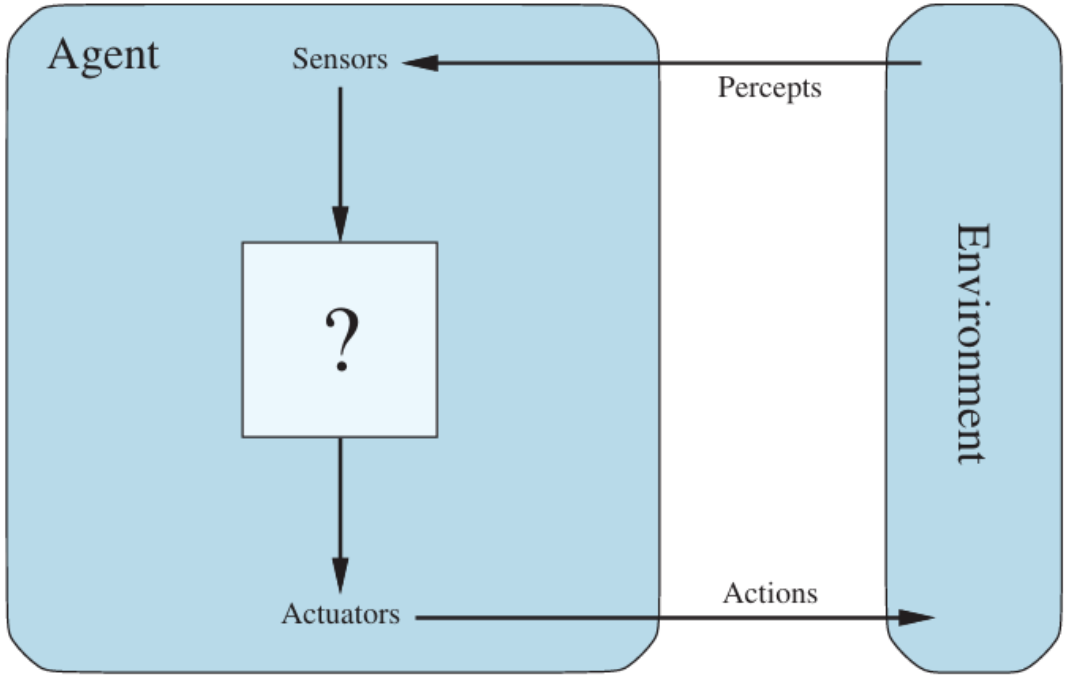


Figure 2.1 Agents interact with environments through sensors and actuators.

KEY COMPONENTS

Sensors (Perception)

- Gather information from the environment
- Monitor changes and inputs
- Enable understanding of surroundings

Actuators (Action)

- Execute actions in the environment
- Influence surroundings based on decisions
- Produce responses and outputs

EXAMPLES OF INTELLIGENT AGENTS

Human Agent

Sensors
- Eyes
- Ears
- Nose
- Skin

Actuators
- Hands
- Legs
- Vocal tract

Robotic Agent

Sensors
- Cameras
- Infrared sensors
- Range finders

Actuators
- Motors
- Wheels
- Robotic arms

Software Agent

Sensors (Inputs)
- Keyboard / Mouse / Touchscreen
- Voice commands
- Network packets
- File contents

Actuators (Outputs)
- Display information
- Generate sounds
- Send network packets
- Write files

PERCEPTS, PERCEPT SEQUENCES, AND AGENT FUNCTIONS

Percept

- A percept is the information an agent receives through its sensors.
- It represents the agent's current observation of the environment.

Percept Sequence

- A percept sequence is the complete history of all percepts received by the agent.
- It contains everything the agent has perceived up to the present moment.

Action Selection

- An agent chooses actions based on:
  - Its built-in knowledge
  - Its accumulated percept sequence
- An agent cannot use information that it has not perceived.

Agent Behavior

- An agent's behavior is determined by how it responds to its percept sequence.
- Different percept sequences may lead to different actions.

Agent Function

- An agent function maps percept sequences to actions.

AGENT FUNCTION VS AGENT PROGRAM

Agent Function

- Describes the behavior of an agent.
- Maps every possible percept sequence to an appropriate action.
- Represents a theoretical or mathematical model of the agent.
- For complex agents, the function may be extremely large or even infinite.

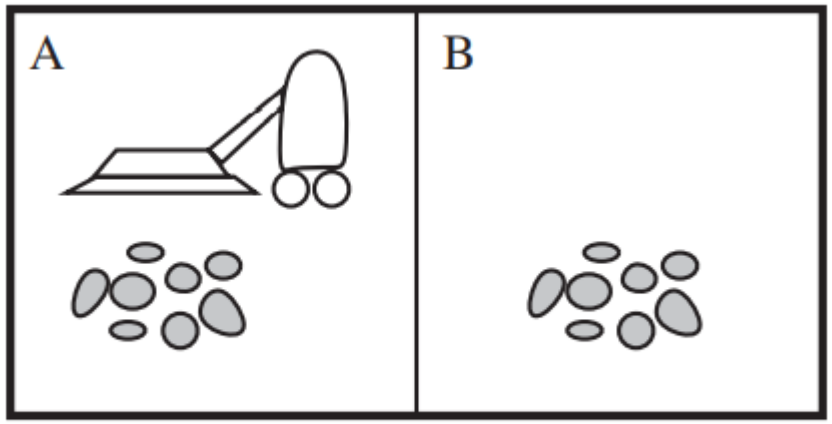
Characteristics

- External description of agent behavior.
- Defines "What action should be taken?"
- Independent of implementation details.

Agent Program

- A concrete implementation of the agent function.
- Runs on a physical system (computer, robot, mobile device, etc.).
- Processes percepts and generates actions. Implements the logic specified by the agent function.

TO ILLUSTRATE THESE IDEAS, WE USE A SIMPLE EXAMPLE—THE VACUUM-CLEANER WORLD, WHICH CONSISTS OF A ROBOTIC VACUUM-CLEANING AGENT IN A WORLD CONSISTING OF SQUARES THAT CAN BE EITHER DIRTY OR CLEAN.



A vacuum-cleaner world with just two locations.

- Vacuum-cleaner world shown in Figure.
- This world has just two locations: squares A and B.
- The vacuum agent perceives which square it is in and whether there is dirt in the square.
- It can choose to move left, move right, suck up the dirt, or do nothing.
- One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.

- A partial tabulation of this agent function is shown in below:

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
⋮	⋮
$[A, Clean], [A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Clean], [A, Dirty]$	$Suck$
⋮	⋮

An agent program that implements it appears as shown below:

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.

GOOD BEHAVIOR: CONCEPT OF RATIONALITY

- A **rational agent** is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly.
- Obviously, doing the right thing is better than doing the wrong thing.

WHAT DOES IT MEAN TO DO THE RIGHT THING?

- Will answer this by considering the consequences of the agent's behavior.
- When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states. If the sequence is desirable, then the agent has performed well. This notion of desirability is captured by a performance measure that evaluates any given sequence of environment states.
- We said environment states, not agent states.
- If we define success in terms of agent's opinion of its own performance, an agent could achieve perfect rationality simply by deluding itself that its performance was perfect.
- Human agents in particular are notorious for "sour grapes"—believing they did not really want something (e.g., a Nobel Prize) after not getting it.

- There is not one fixed performance measure for all tasks and agents.
- How to measure performance of rational agent?
    - ➢ Give reward for every right action.
    - ➢ Don’t give reward for wrong action.
    - ➢Total reward is used to measure the performance.
- A general rule, it is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.

RATIONALITY

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

This leads to a definition of a rational agent:

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

- Consider the simple vacuum-cleaner agent that cleans a square if it is dirty and moves to the other square if not;
- this is the agent function tabulated in Figure 2.3.
- Is this a rational agent?
- That depends! First, we need to say what the performance measure is, what is known about the environment, and what sensors and actuators the agent has. Let us assume the following:
  - ✓ The performance measure awards one point for each clean square at each time step, over a "lifetime" of 1000 time steps.
  - ✓ The "geography" of the environment is known a priori (Figure 2.2) but the dirt distribution and the initial location of the agent are not. Clean squares stay clean and sucking cleans the current square. The Right and Left actions move the agent one square except when this would take the agent outside the environment, in which case the agent remains where it is.
  - ✓ The only available actions are Right, Left, and Suck.
  - ✓ The agent correctly perceives its location and whether that location contains dirt.
- Under these circumstances the agent is indeed rational; its expected performance is at least as good as any other agent's.

OMNISCIENCE, LEARNING, AND AUTONOMY

- We need to be careful to distinguish between rationality and omniscience.
- An omniscient agent knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.
- Consider the following example:

I am walking along the Champs Elysees one day and I see an old friend across the street. There is no traffic nearby and I'm not otherwise engaged, so, being rational, I start to cross the street. Meanwhile, at 33,000 feet, a cargo door falls off a passing airliner,3 and before I make it to the other side of the street I am flattened.

- Was I irrational to cross the street?
- This example shows that rationality is not the same as perfection.
- Rationality maximizes expected performance, while perfection maximizes actual performance.
- Doing actions in order to modify future percepts—sometimes called **information gathering**
- a rational agent not only to gather information but also to **learn** as much as possible from what it perceives.

- To the extent that an agent relies on the prior knowledge of its designer rather than on its own percept's and learning processes, we say that the agent lacks autonomy.
- A rational agent should be autonomous—it should learn what it can to compensate for partial or incorrect prior knowledge.
- For example, a vacuum-cleaning agent that learns to predict where and when additional dirt will appear will do better than one that does not.

THE NATURE OF ENVIRONMENTS

- we must think about task environments, which are essentially the “problems” to which rational agents are the “solutions.”
- show that task environments come in a variety of flavors.
- The nature of the task environment directly affects the appropriate design for the agent program.

SPECIFYING THE TASK ENVIRONMENT

- PEAS (Performance, Environment, Actuators, Sensors) description.
- In designing an agent, the first step must always be to specify the task environment as fully as possible.
- Figure 2.4 summarizes the PEAS description for the taxi's task environment.
- We discuss each element in more detail in the following paragraphs.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

WHAT IS THE PERFORMANCE MEASURE?

- ✓ Getting to the correct destination;
- ✓ Minimizing fuel consumption and wear and tear;
- ✓ Minimizing the trip time or cost;
- ✓ Minimizing violations of traffic laws and disturbances to other drivers;
- ✓ Maximizing safety and passenger comfort;
- ✓ Maximizing profits.

what is the driving environment that the taxi will face?

- Deal with a variety of roads, ranging from rural lanes and urban alleys to 12-lane freeways.
- The roads contain other traffic, pedestrians, stray animals, road works, police cars, puddles, and potholes.
- The taxi must also interact with potential and actual passengers.

The **actuators** for an automated taxi include those available to a human driver:

- control over the engine through the accelerator and control over steering and braking.

The basic **sensors** for the taxi will include:

- one or more controllable video cameras so that it can see the road; it might augment these with infrared or sonar sensors to detect distances to other cars and obstacles.
- To avoid speeding tickets, the taxi should have a speedometer, and to control the vehicle properly, especially on curves, it should have an accelerometer.
- To determine the mechanical state of the vehicle, it will need the usual array of engine, fuel, and electrical system sensors.
- Like many human drivers, it might want a global positioning system (GPS) so that it doesn’t get lost.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

ENVIRONMENT

- An environment is everything in the world which surrounds the agent, but it is not a part of an agent itself.
- An environment can be described as a situation in which an agent is present.

PROPERTIES OF TASK ENVIRONMENTS

1. Fully observable vs Partially Observable
2. Single-agent vs Multi-agent
3. Deterministic vs Stochastic
4. Episodic vs sequential
5. Static vs Dynamic
6. Discrete vs Continuous
7. Known vs Unknown
8. Accessible vs Inaccessible

1. FULLY OBSERVABLE VS PARTIALLY OBSERVABLE

- If an agent sensor can sense or access the complete state of an environment at each point of time then it is a **fully observable** environment, else it is **partially observable.**
- A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
- When sensing its environment, **an agent often receives information that only partially describes the current state of affairs**.
- The agent then attempts to predict what it has not sensed, by using other pieces of information available through its sensors.

EXAMPLE

- Example for Fully Observable:

In a chess game, the state of the system, that is, the position of all the players on the chess board, is available the whole time so the player can make an optimal decision.

- Example for Partially Observable:

A vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking.

2. SINGLE-AGENT VS MULTI-AGENT

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.

Examples:

- An agent solving a crossword puzzle by itself is clearly in a single-agent environment.
- Playing a soccer match is a multi-agent environment.

- we have described how an entity may be viewed as an agent, but we have not explained which entities must be viewed as agents.
- Does an agent A (the taxi driver for example) have to treat an object B (another vehicle) as an agent?

B's behavior is best described as maximizing a performance measure whose value depends on agent A's behavior.

For example:

- In chess, the opponent entity B is trying to maximize its performance measure, which, by the rules of chess, minimizes agent A's performance measure. Thus, chess is a competitive multiagent environment.
- In the taxi-driving environment, on the other hand, avoiding collisions maximizes the performance measure of all agents, so it is a partially **cooperative** multiagent environment.

3. DETERMINISTIC VS STOCHASTIC

- If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.
- A stochastic environment is random in nature and cannot be determined completely by an agent.

Examples:

- The traffic signal is a deterministic environment where the next signal is known for a pedestrian (**Agent**).
- The radio station is a stochastic environment where the listener is not aware about the next song or playing a soccer is stochastic environment.

4. EPISODIC VS SEQUENTIAL

- **Episodic** is an environment where each state is independent of each other. The action on a state has nothing to do with the next state.

Example:

Suppose A bot (agent) answer to a question and then answer to another question and so on. So, each question-answer is a single episode.

- **The sequential** environment is an environment where the next state is dependent on the current action. So, agent current action can change all of the future states of the environment.

Example:

- Playing tennis is a perfect example where a player observes the opponent's shot and takes action.

5. STATIC VS DYNAMIC

- If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
- Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
- However for dynamic environment, agents need to keep looking at the world at each action.
- Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.

Example: Cleaning a room (Environment) by a dry-cleaner (Agent) is an example of a static environment where the room is static while cleaning.
Example: Playing soccer is a dynamic environment where players' positions keep changing throughout the game. So a player hit the ball by observing the opposite team.

6. DISCRETE VS CONTINUOUS

- **Discrete Environment** consists of a finite number of states and agents have a finite number of actions.

- Real-life Example: Choices of a move (action) in a tic-tac game are finite on a finite number of boxes on the board (Environment).
- A chess gamecomes under discrete environment as there is a finite number of moves that can be performed.
- While in a **Continuous environment**, the environment can have an infinite number of states. So, the possibilities of taking an action are also infinite.

- Real-life Example: In a basketball game, the position of players (**Environment**) keeps changing continuously and hitting (**Action**) the ball towards the basket can have different angles and speed so infinite possibilities.
- A self-driving car is an example of a continuous environment.

7. KNOWN VS UNKNOWN

- Known and unknown are not actually a feature of an environment, but it is an agent's state of knowledge to perform an action.
- In a known environment, the results for all actions are known to the agent. While in unknown environment, agent needs to learn how it works in order to perform an action.
- It is quite possible that a known environment to be partially observable and an Unknown environment to be fully observable.

8. ACCESSIBLE VS INACCESSIBLE

- If an agent can obtain complete and accurate information about the state's environment, then such an environment is called an Accessible environment else it is called inaccessible.
- An empty room whose state can be defined by its temperature is an example of an accessible environment.
- Information about an event on earth is an example of Inaccessible environment.

THE STRUCTURE OF AGENTS

- The job of AI is to design an agent program that implements the agent function— the mapping from percepts to actions.
- This program will run on some sort of computing device with physical sensors and actuators—we call this the architecture:

agent = architecture + program

- Program we choose has to be one that is appropriate for the architecture.
- If the program is going to recommend actions like Walk, the architecture had better have legs.
- The architecture might be just an ordinary PC, or it might be a robotic car with several onboard computers, cameras, and other sensors.
- In general, the architecture makes the percepts from the sensors available to the program, runs the program, and feeds the program's action choices to the actuators as they are generated.

AGENT PROGRAMS

- Take the current percept as input from the sensors and return an action to the actuators.
- **Agent program is an implementation of an agent function**.
- An agent function is a map from the percept sequence(history of all that an agent has perceived to date) to an action.

Designers must construct a table that contains the appropriate action for every possible percept sequence.

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

Figure 2.7 The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

Four basic kinds of agent programs that embody the principles underlying almost all intelligent systems:

1. Simple reflex agents
2. Model-based reflex agents
3. Goal-based agents
4. Utility-based agents

Each kind of agent program combines particular components in particular ways to generate actions.

1. SIMPLE REFLEX AGENTS

- A simple reflex agent is an AI agent that selects actions based on its current perception of the world and not based on past perceptions.
- It does this through pre-determined rules, also known as condition-action rules.
- If the condition is true, then the action is taken.

Example: the vacuum agent whose agent function is tabulated in Figure 2.3 is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt.

- If the current perception is "dirt," then the action is "suction on."
- If the current perception is "no dirt," then the action is "move forward."

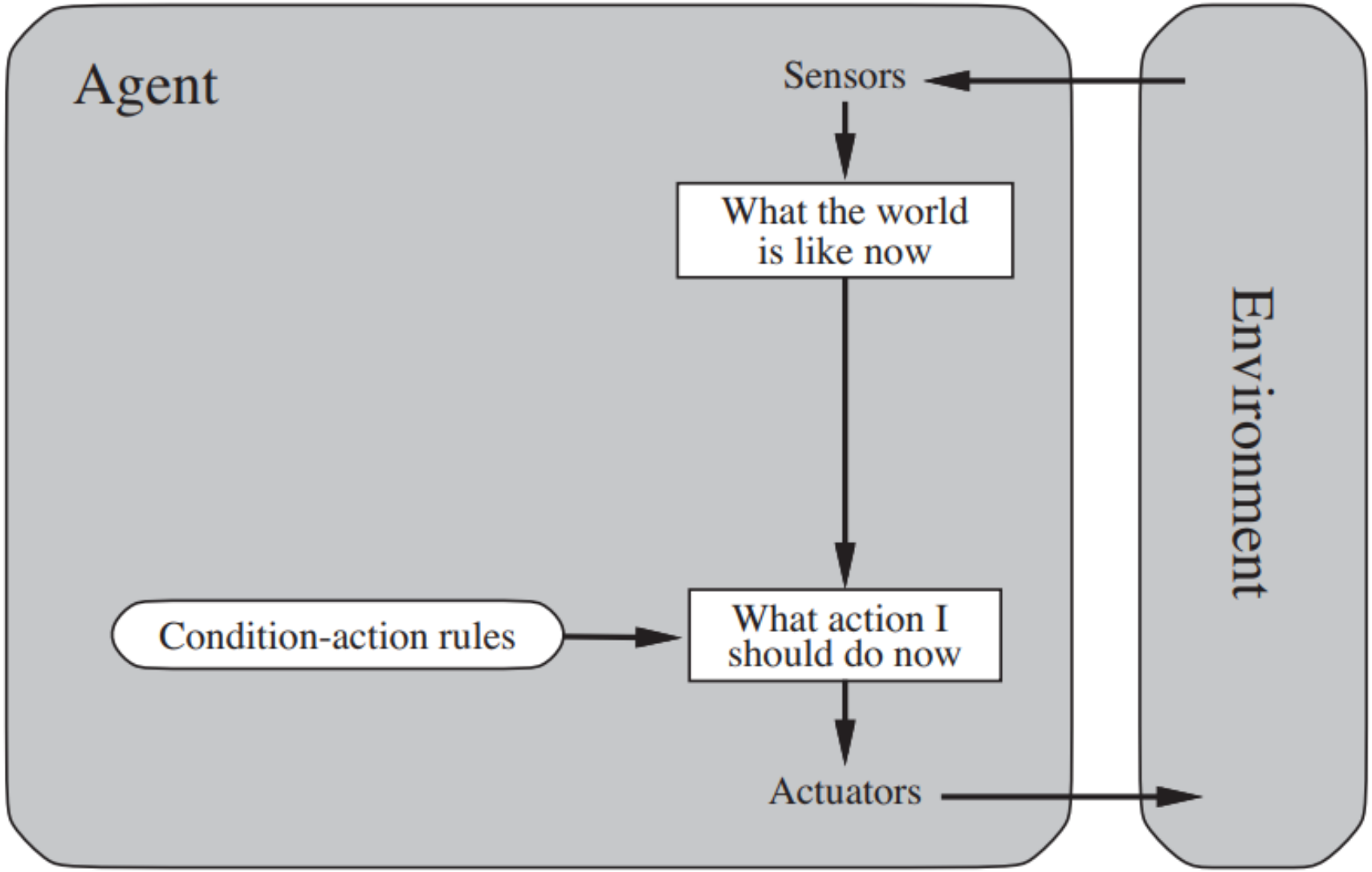
HOW A SIMPLE REFLEX AGENT WORKS?

- The agent perceives its current state by observing its environment.
- The agent matches its current state to a condition-action rule.
- The agent takes the action that is specified by the rule.

The agent repeats this process until it terminates.

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.



.9 Schematic diagram of a simple reflex agent.

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.10 A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

- Simple reflex agents are the simplest type of AI agent.
- They are easy to understand and implement. However, they are also quite limited in their capabilities.
- They cannot learn from their experiences and they cannot handle complex environments.

2. MODEL-BASED REFLEX AGENTS

- Uses internal memory and a percept history to create a model of the environment in which it's operating and make decisions based on that model.
- A model-based reflex agent is AI agent that uses a model of the world to make decisions.
- The model is a representation of the agent's environment and how it is likely to change over time.
- The agent uses the model to predict the consequences of its actions and to choose the action that is most likely to achieve its goal.
- Model-based reflex agents are more complex than simple reflex agents, which do not use a model of the world. However, model-based reflex agents are able to make better decisions in complex environments, especially when the environment is partially observable or when the agent needs to plan ahead.

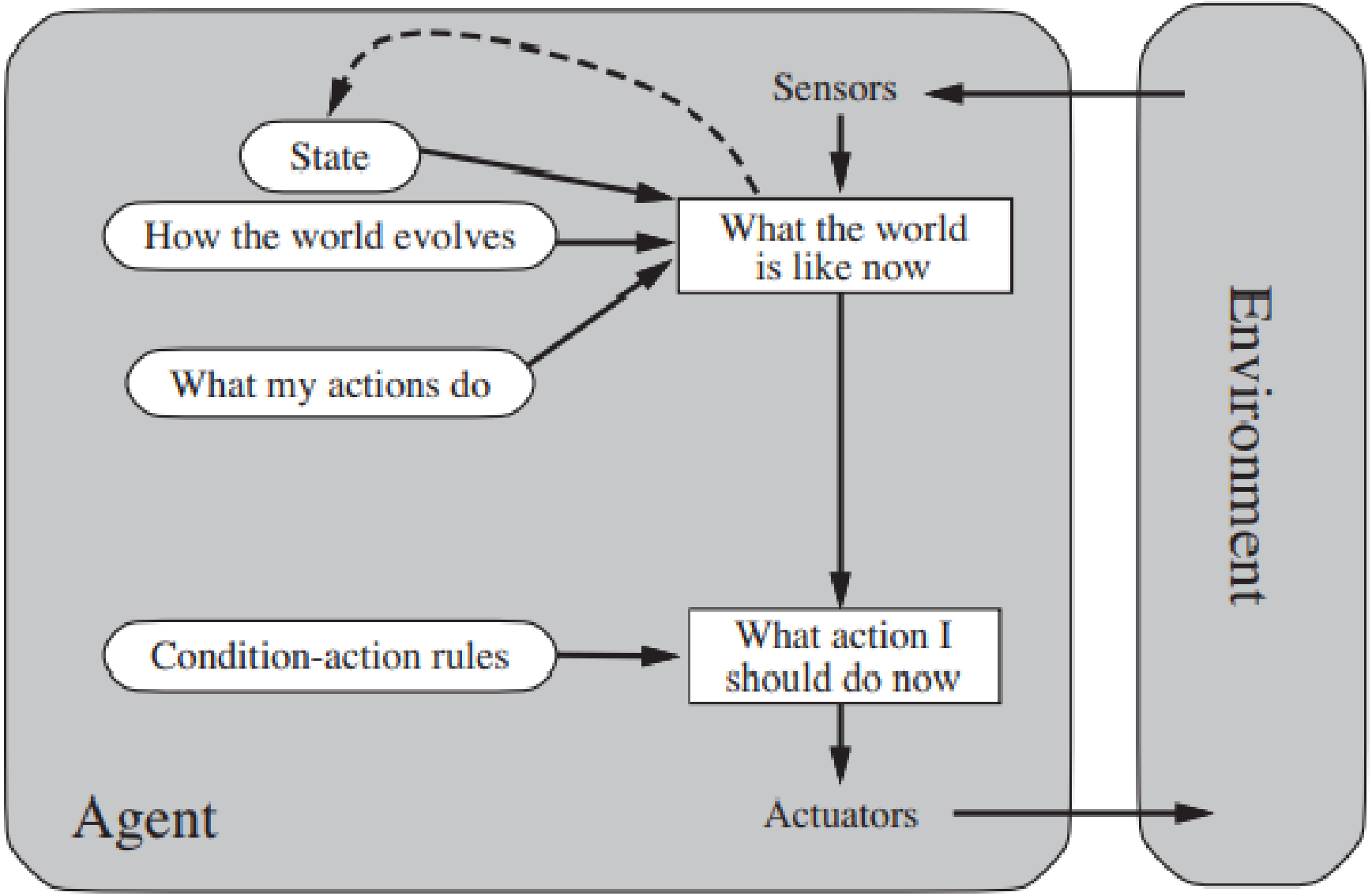
Example: **Self-driving cars** are a great example of a model-based reflex agent. The car is equipped with sensors that detect obstacles, such as car brake lights in front of them or pedestrians walking on the sidewalk. As it drives, these sensors feed percepts into the car's memory and internal model of its environment.

- Model-based reflex agents are used in a variety of AI applications, such as robotics, self-driving cars, and video games.
- For example, a model-based reflex agent could be used to control a robot that is navigating a maze. The agent would use its model of the maze to predict where it can go and where it should go to reach its goal.

HOW A MODEL-BASED REFLEX AGENT WORKS?

- The agent perceives its current state by observing its environment.
- The agent uses its model of the world to predict the consequences of its actions.
- The agent chooses the action that is most likely to achieve its goal.
- The agent takes the action and observes the new state of the world.
- The agent updates its model of the world to reflect the new state.
- The agent repeats this process until it reaches its goal or until it terminates.

- Model-based reflex agents are a powerful tool for developing intelligent agents.
- However, it is important to note that model-based reflex agents can be computationally expensive and that they can be difficult to implement in real-world environments.



2.11 A model-based reflex agent.

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              model, a description of how the next state depends on current state and action
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.12 A model-based reflex agent. It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

3. GOAL-BASED AGENTS

- Extension of model-based agents.
- A goal-based agent is an AI agent that has a goal and selects actions in order to achieve that goal.
- It uses a model of the world and a plan to achieve its goal.
- The agent's plan is a sequence of actions that the agent believes will lead it to its goal.
- Goal-based agents are able to make better decisions in complex environments, especially when the agent needs to plan ahead.

Examples:

- A robot that is trying to navigate a maze.
- A self-driving car that is trying to get its passengers to their destination safely.
- A video game character that is trying to complete a level.
- A personal assistant that is trying to schedule appointments and make reservations for its user.

HOW A GOAL-BASED AGENT WORKS?

- The agent sets a goal for itself.
- The agent creates a plan to achieve its goal.
- The agent executes its plan.
- The agent monitors its progress and updates its plan as needed.
- The agent repeats this process until it reaches its goal or until it terminates.

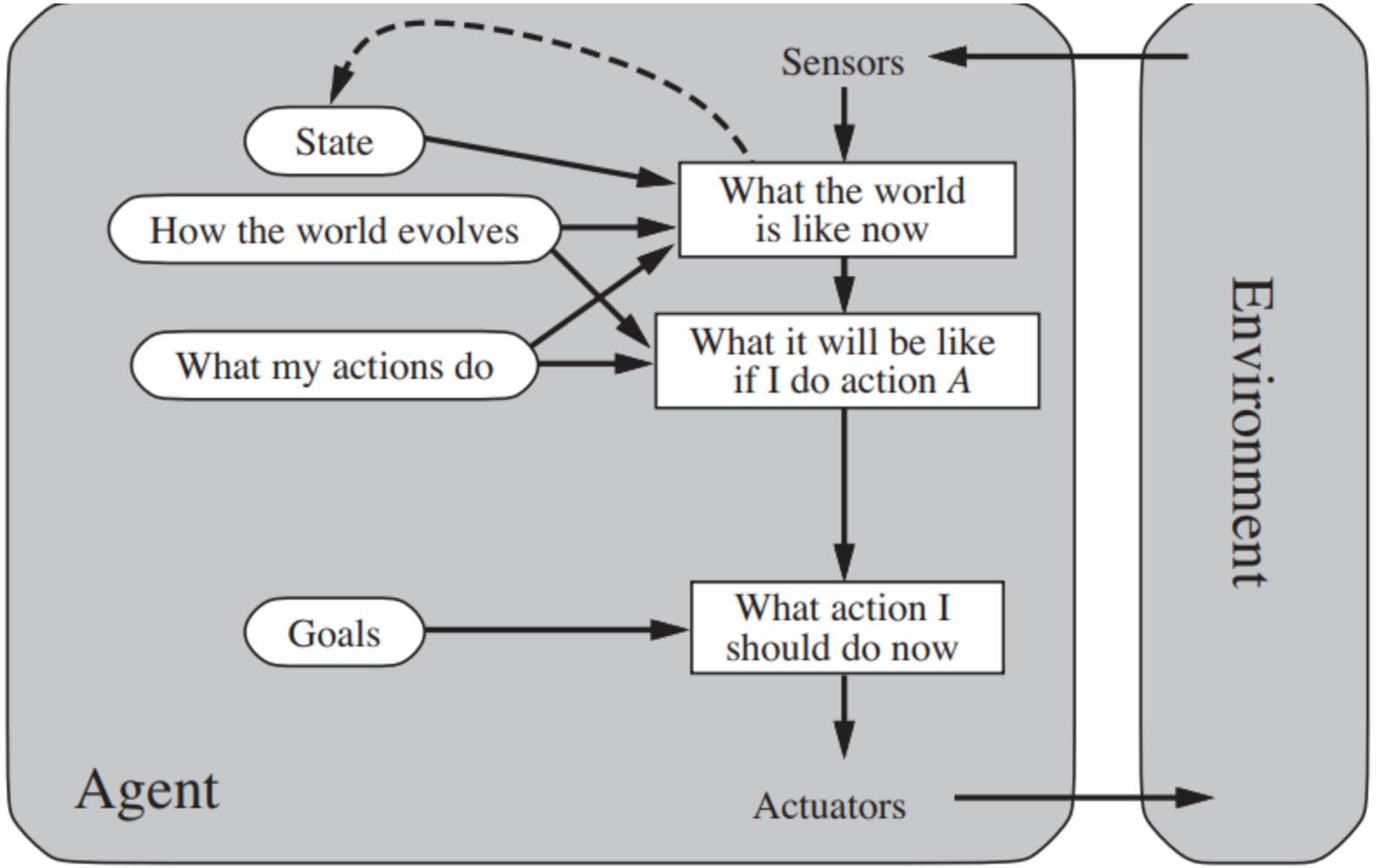


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

4. UTILITY-BASED AGENTS

- A utility-based agent is **an agent that acts based not only on what the goal is, but the best way to reach that goal**.
- A goal-based navigation agent is tasked with getting from point A to point B.
- If the agent succeeds, the goal has been satisfied.
- A utility-based navigation agent could seek to get from point A to point B in the shortest amount of time, with the minimum expenditure of fuel, or both.
- A utility-based agent is an AI agent that makes decisions based on a utility function.
- A utility function is a mathematical function that maps states of the world to real numbers, representing how desirable or undesirable those states are to the agent.
- The agent's goal is to maximize its expected utility, which is the expected value of the utility function over all possible states of the world that the agent could end up in.

- More flexible than goal-based agents, which are a simpler type of AI agent that only has one goal and makes decisions based on whether or not an action will achieve that goal.
- Can take into account multiple factors, such as the cost of achieving a goal, the risks involved, and the agent's own preferences.
- This makes them more suitable for complex environments, where there may be multiple possible goals and the agent needs to weigh different factors to decide which action to take.

For example:

- A robotic vacuum cleaner might have a utility function that rewards it for cleaning up dirt and debris, but penalizes it for bumping into furniture or running out of battery. The robot would then use this utility function to decide how to move around the environment and clean as efficiently as possible.
- A utility-based agent is a stock trading agent. The utility function for this agent might reward it for buying stocks that go up in value, but penalize it for buying stocks that go down in value. The agent would then use this utility function to decide which stocks to buy and sell, in order to maximize its expected profits.

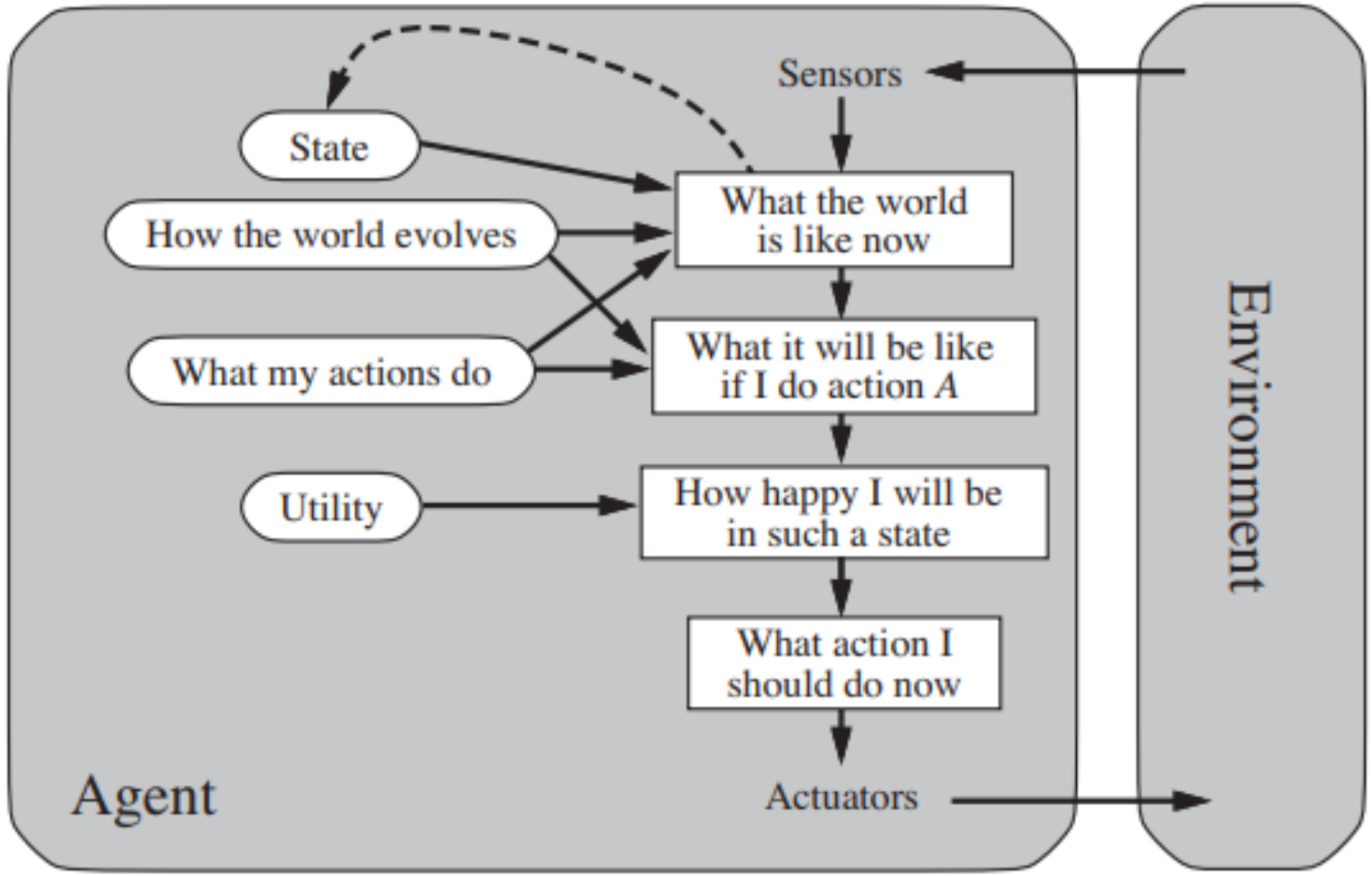


Figure 2.14 A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

LEARNING AGENTS

- A learning agent can improve its performance over time by learning from experience. Learning agents are typically equipped with a set of learning algorithms, which allow them to update their knowledge and behavior based on the feedback they receive from the environment.
- Learning allows the agent to operate in initially unknown environments and to become more competent than its initial knowledge alone might allow.

Learning element:

- Which is responsible for making improvements.
- The learning element uses CRITIC feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.

Performance element:

- which is responsible for selecting external actions.
- The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions.

Critic:

- The critic tells the learning element how well the agent is doing with respect to a fixed performance standard.

Problem generator:

- It is responsible for suggesting actions that will lead to new and informative experiences.

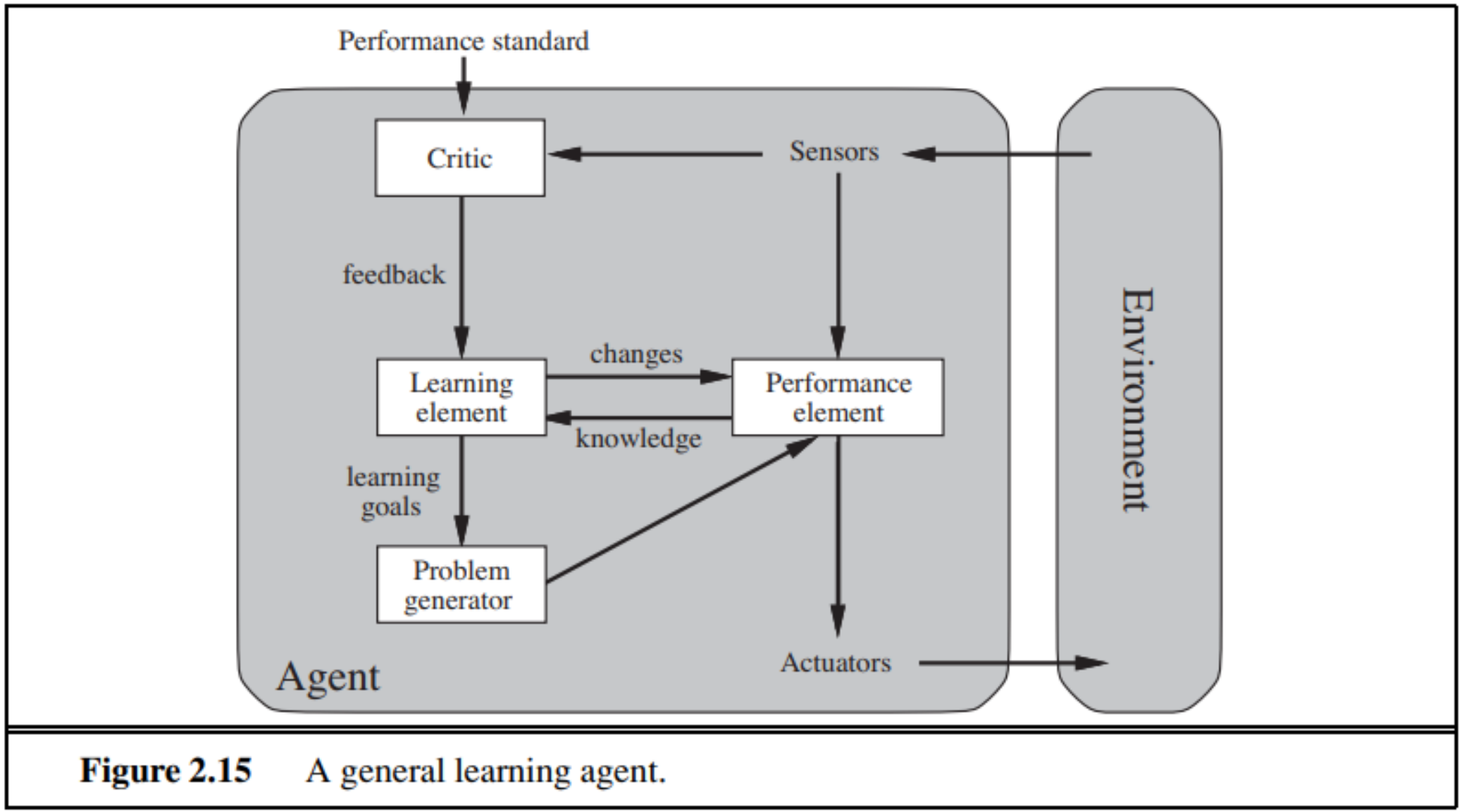


Figure 2.15 A general learning agent.

Automated taxi example:

- The **performance element** consists of whatever collection of knowledge and procedures the taxi has for selecting its driving actions. The taxi goes out on the road and drives, using this performance element.
- The **critic** observes the world and passes information along to the learning element. For example, after the taxi makes a quick left turn across three lanes of traffic, the critic observes the shocking language used by other drivers. From this experience, the learning element is able to formulate a rule saying this was a bad action, and the performance element is modified by installation of the new rule.
- The problem generator might identify certain areas of behavior in need of improvement and suggest experiments, such as trying out the brakes on different road surfaces under different conditions.

Different types of learning algorithms, but some of the most common include:

- **Reinforcement learning:** Reinforcement learning agents learn by trial and error. They are rewarded for taking actions that lead to desirable outcomes, and penalized for taking actions that lead to undesirable outcomes. Over time, the agent learns to take the actions that are most likely to lead to a high reward.
- **Supervised learning:** Supervised learning agents learn from examples. They are given a set of training examples, which consist of pairs of inputs and outputs. The agent learns to predict the output for a given input by analyzing the training examples.
- **Unsupervised learning:** Unsupervised learning agents learn from data without any labels. They try to identify patterns and relationships in the data, without being told what to look for.

For example:

- A self-driving car might use a reinforcement learning algorithm to learn how to drive safely and efficiently.
- A game playing agent might use a supervised learning algorithm to learn how to play a game optimally.
- And a natural language processing agent might use an unsupervised learning algorithm to learn the meaning of words and phrases.

END of Unit-1